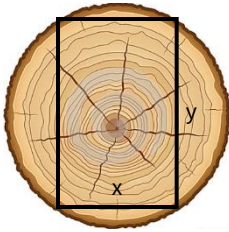
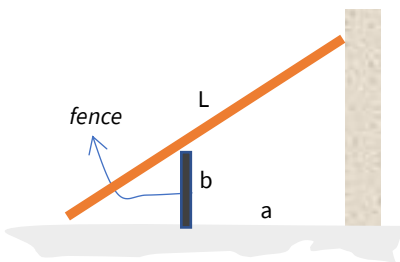


**Variables in relation to Maxima/Minima Values – continuation...****Stiffest beam that can be cut from a circular log of radius r .**Stiffness is proportional to the product of the breadth, x and cube of the width, y .

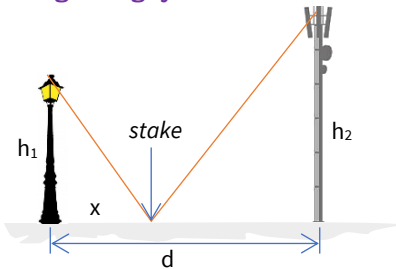
$$y = x\sqrt{3}$$

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Location of single stake at ground level to minimize the length of guy wire.

$$L = \sqrt[3]{(a^{2/3} + b^{2/3})^2}$$

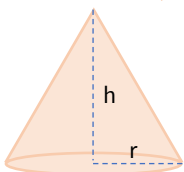
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Location of single stake at ground level to minimize the length of guy wire.

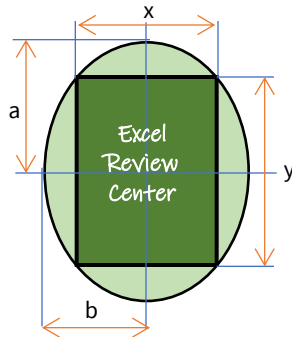
$$x = \frac{dh_1}{h_1 + h_2}$$

Least amount of material for cone

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$$r = \frac{h}{\sqrt{2}}$$

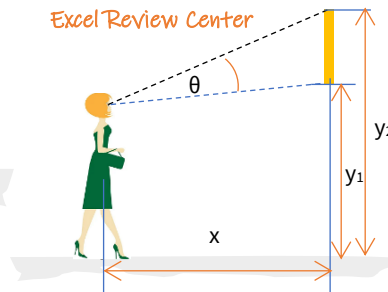
Strongest beam that can be cut from an elliptical section.Strength is proportional to the product of breadth, x and the square of the depth, y .

$$x = 2b\sqrt{\frac{1}{3}}$$

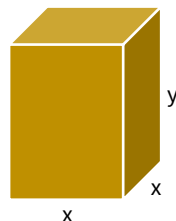
$$y = 2a\sqrt{\frac{2}{3}}$$

Best possible view of picture or clock.

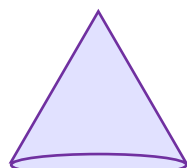
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Best view means θ is maximum.

$$x = \sqrt{y_1 y_2}$$

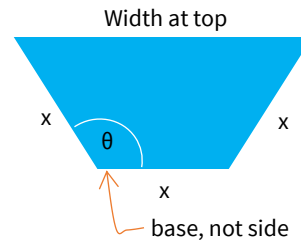
Least amount of material for square-based parallelepiped

$$x = 2y$$
$$x = \sqrt[3]{2V}$$
$$V = \text{volume}$$

Maximum volume of cone with given slant height, s .

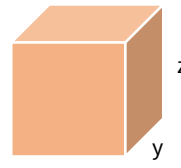
$$h = \frac{s}{\sqrt{3}}$$
$$\theta = \tan^{-1} \sqrt{2}$$

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Most efficient trapezoidal section (Maximum capacity with minimum perimeter)

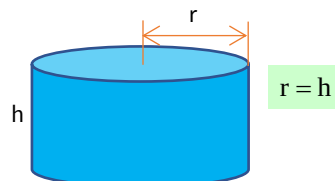
It is half of a regular hexagon

$$\text{width at top} = \text{sum of sides}$$
$$\text{width at top} = 2x$$
$$\theta = 120^\circ$$

Parallelepiped with maximum volume

The parallelepiped is a cube.

$$x = y = z$$

Least amount of material for an open top cylindrical tank

$$r = h$$

Maximum volume of right cylinder inscribed in a sphere of radius R

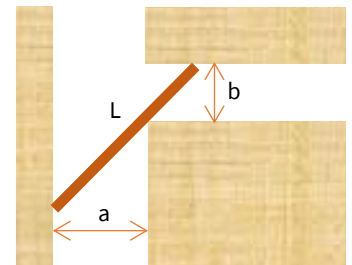
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$$V_c = \frac{4}{\sqrt{27}} \pi R^3$$
$$h = \frac{2\sqrt{3}}{3} R$$

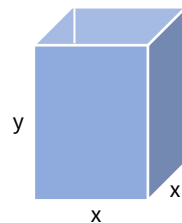
Volume of largest cone inscribed in a hemisphere

$$V_c = \frac{V_H}{2}$$

Maximum length of rigid beam that can pass perpendicular hallways

$$L = \sqrt[3]{(a^{2/3} + b^{2/3})^2}$$

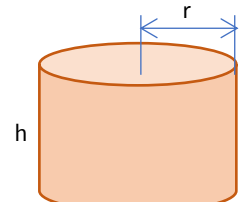
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Open square based container with maximum volume.

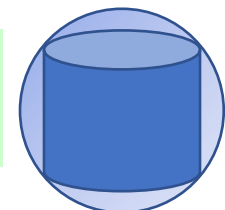
$$x = 2y$$

$$x = \sqrt{\frac{SA}{3}}$$

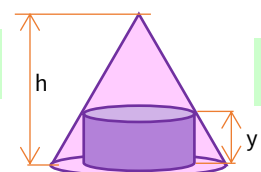
SA = surface area

Minimum cost of materials for cylinder with a given volume.

$$r = \sqrt[3]{\frac{V}{2\pi}}$$

Ratio of the weight of heaviest cylinder to weight of circumscribing sphere

$$\frac{W_c}{W_s} = \frac{1}{\sqrt{3}}$$

Largest cylinder inscribed in a cone.

$$y = \frac{h}{3}$$